

LISBOA PORTELLA AP, Portugal

WMO: 085360

Lat: 38.781N

Lon: 9.136W

Elev: 114

StdP: 99.96

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	6.0	-2.9	3.0	10.1	-0.9	3.6	11.0	11.3	13.2	10.2	13.4	3.1	50	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.2	34.1	19.5	32.1	19.2	30.1	19.0	21.7	28.2	21.1	27.1	20.6	26.2	4.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.1	22.0	19.8	14.7	21.9	19.0	14.0	21.7	63.8	28.4	61.5	27.4	59.8	26.3	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.4	8.4	DB	2.3	38.3	1.5	2.1	1.2	39.9	0.3	41.1	-0.5	42.3	-1.6	43.9	(4)
(5)			WB	0.7	23.0	1.5	0.6	-0.4	23.4	-1.3	23.8	-2.1	24.1	-3.2	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	11.6	12.3	14.5	16.0	18.4	21.5	23.1	23.9	22.3	19.3	14.7	12.1	(6)	
	DBStd	5.08	2.37	2.22	2.64	2.91	3.12	3.09	2.75	2.66	2.41	2.51	2.50	2.45	(7)	
	HDD10.0	28	12	6	1	0	0	0	0	0	0	0	1	8	(8)	
	HDD18.3	947	209	170	123	83	36	4	0	0	0	18	112	192	(9)	
	CDD10.0	2764	61	69	140	179	262	345	407	431	368	287	141	74	(10)	
	CDD18.3	642	0	0	3	13	39	99	148	172	118	47	2	0	(11)	
	CDH23.3	4272	0	0	22	78	282	706	1081	1268	655	179	1	0	(12)	
(13)	CDH26.7	1550	0	0	1	12	89	273	425	504	209	37	0	0	(13)	
(14)	Wind	WSAvg	4.2	3.6	3.9	4.1	4.2	4.5	4.6	5.1	4.8	4.1	3.7	3.6	3.6	(14)
(15) Precipitation	PrecAvg	735	105	92	72	62	47	18	3	5	30	94	121	101	(15)	
	PrecMax	1189	353	298	267	163	156	98	43	58	129	262	373	383	(16)	
	PrecMin	416	0	0	1	5	0	0	0	0	0	0	2	4	(17)	
	PrecStd	206	76	67	59	40	22	7	11	32	74	86	9	1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	20.1	25.9	28.2	32.2	35.2	36.9	37.1	33.9	30.1	23.0	18.8	(19)	
		MCWB	15.0	14.4	15.0	17.1	18.9	20.1	20.1	19.9	19.3	17.6	16.6	16.5	(20)	
	2%	DB	17.0	18.1	23.1	25.8	29.1	32.1	33.9	34.1	31.2	27.1	21.0	17.9	(21)	
		MCWB	14.7	13.7	14.5	16.4	17.9	19.4	19.6	19.5	18.7	17.5	17.6	16.5	(22)	
	5%	DB	16.1	17.0	21.1	23.2	26.2	29.9	31.0	31.8	29.1	25.1	19.8	17.0	(23)	
		MCWB	14.4	13.3	14.1	15.5	17.3	19.0	19.5	19.3	18.6	17.5	16.8	15.6	(24)	
(25) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	15.2	16.0	19.0	21.0	24.0	27.2	28.8	29.1	27.0	23.2	18.8	16.1	(25)	
		MCWB	13.5	13.4	13.7	15.1	16.7	18.4	19.2	19.1	18.3	17.4	16.2	14.6	(26)	
	0.4%	WB	16.9	16.2	17.2	19.2	21.0	21.7	22.2	22.6	21.9	21.5	20.1	18.1	(27)	
		MCDB	17.2	17.6	21.4	25.9	28.5	30.7	31.5	30.4	26.5	23.6	21.2	18.2	(28)	
	2%	WB	15.8	15.5	16.1	17.5	19.5	20.8	21.2	21.7	21.1	20.5	19.0	17.1	(29)	
		MCDB	16.4	16.9	19.2	23.1	25.5	28.6	29.1	28.1	25.4	22.9	19.8	17.5	(30)	
(31) Mean Daily Temperature Range	5%	WB	15.0	14.8	15.4	16.5	18.4	19.9	20.6	21.1	20.5	19.7	17.8	16.0	(31)	
		MCDB	15.8	16.2	18.4	21.1	23.7	27.0	27.7	27.0	24.6	22.0	18.9	16.7	(32)	
	10%	WB	14.1	14.0	14.7	15.6	17.5	19.1	19.9	20.4	19.8	19.1	16.6	14.9	(33)	
		MCDB	15.1	15.4	17.8	19.4	22.3	25.3	26.3	26.0	23.8	21.3	18.0	16.0	(34)	
	5% DB	MDBR	6.0	6.8	7.7	7.9	8.5	9.4	9.9	10.2	9.1	7.5	6.2	5.8	(35)	
		MCDBR	6.3	8.1	11.3	11.4	12.7	13.5	13.8	13.6	12.5	10.5	7.1	5.6	(36)	
(37) Clear-Sky Solar Irradiance	taud	MCWBR	4.5	4.9	5.4	5.6	5.8	5.3	4.9	4.9	4.6	4.5	4.4	4.4	(37)	
		MCDBR	4.9	6.3	8.6	9.7	10.2	11.4	11.3	10.5	9.1	6.6	5.5	4.8	(38)	
	Edh at Noon	MCWBR	4.4	4.6	4.8	5.3	5.4	5.0	4.6	4.5	4.2	3.9	4.3	4.5	(39)	
		MCDBR	4.4	4.6	4.8	5.3	5.4	5.0	4.6	4.5	4.2	3.9	4.3	4.5	(39)	
(40) All-Sky Solar Radiation	taub	RadAvg	2.24	3.24	4.40	5.66	6.79	7.33	7.47	6.78	5.30	3.65	2.49	2.02	(44)	
		RadStd	0.28	0.38	0.47	0.46	0.45	0.38	0.30	0.26	0.34	0.30	0.31	0.23	(45)	
	Edh at Noon	RadAvg	2.24	3.24	4.40	5.66	6.79	7.33	7.47	6.78	5.30	3.65	2.49	2.02	(44)	
		RadStd	0.28	0.38	0.47	0.46	0.45	0.38	0.30	0.26	0.34	0.30	0.31	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	-0.30	N/A	N/A	N/A	N/A	-0.45	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page